ECDLP follow-ups: factor-base sweep, pencil pilot and Kummer gate

Reproducible experiment report

21 October 2026

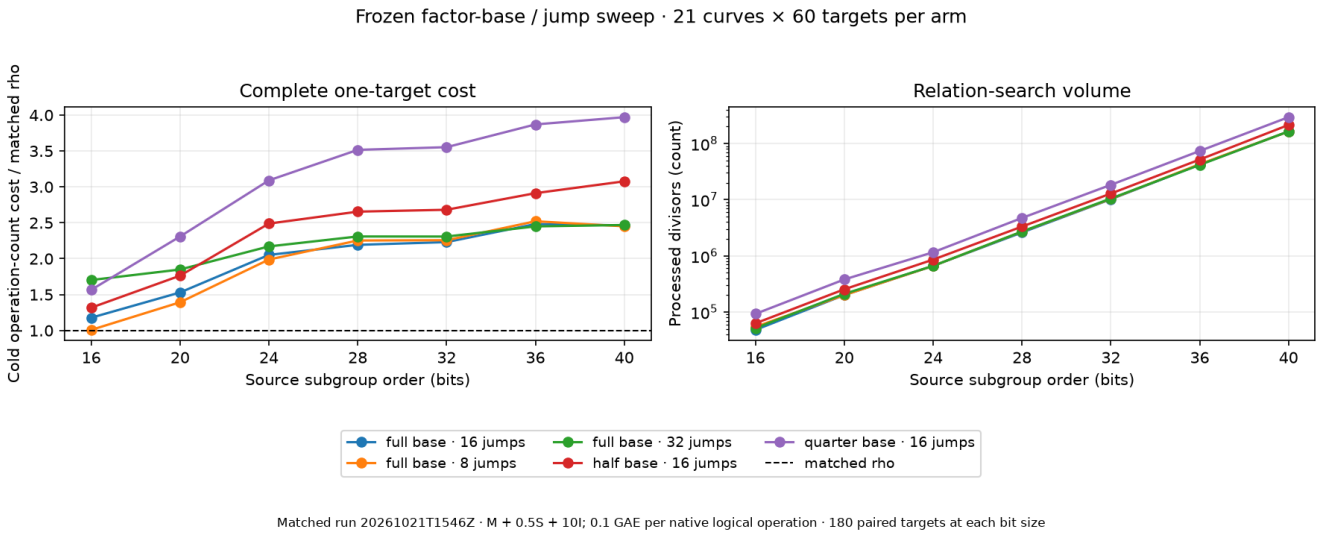
Scope. The frozen protocol covers four leads on 21 exact-order Scholten models, 60 separate targets per model. This report contains completed factor-base and jump experiments, a 540-target pencil pilot, checked branches of the published complete law, and an exact Kummer applicability gate. The full inversion-free law and full-cohort pencil collector remain open. All costs are cold one-target operation-count prices, not wall-speed claims.

Factor-base and jump sweep

Five arms solved and source-verified all 1,260 targets each, with zero failures or cap exhaustions. All 6,300 same-process rho controls were also source-verified. All 1,260 incumbent count and phase records replay exactly. At $M + 0.5S + 10I$ and 0.1 source-addition equivalents (GAE) per declared native logical operation, matched negation rho totals 1,088,802,370 GAE.

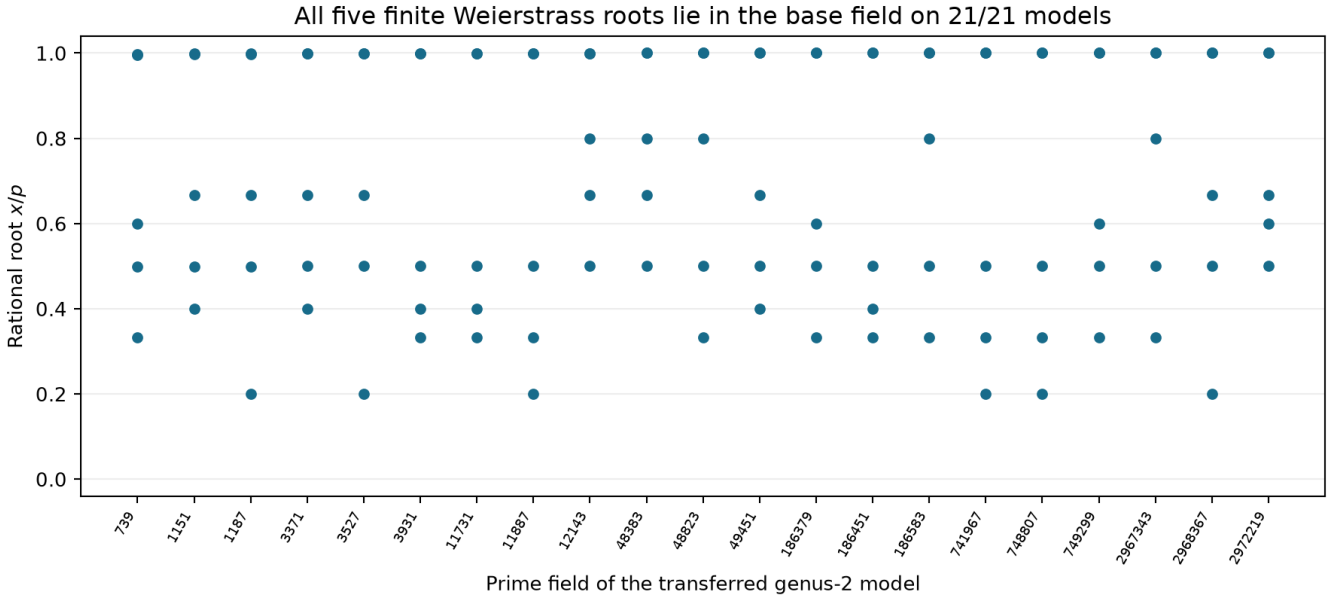
Arm	Divisors processed	Total GAE	/ external	/ rho
Full base, 16 jumps	219,581,773	2,666,291,894	1.000000	2.448830
Full base, 8 jumps	220,092,171	2,664,487,778	0.999323	2.447173
Full base, 32 jumps	219,864,588	2,671,135,032	1.001816	2.453278
Half base, 16 jumps	282,983,105	3,285,713,695	1.232316	3.017732
Quarter base, 16 jumps	392,583,830	4,267,226,320	1.600435	3.919193

The half and quarter bases rejected 35,385,164 and 110,402,348 split relations, respectively, when neither root was in the fixed target-blind base; all rejected candidates were charged. None of the five arms is below rho in the 36 pooled pricing cells. The 8-jump arm is below the 16-jump arm in 15 of 36 cells. Across bit sizes it is below at 16, 20, 24 and 40 bits, and above at 28, 32 and 36 bits. One seed was used per target; no uncertainty interval or isolated wall-time result was measured.

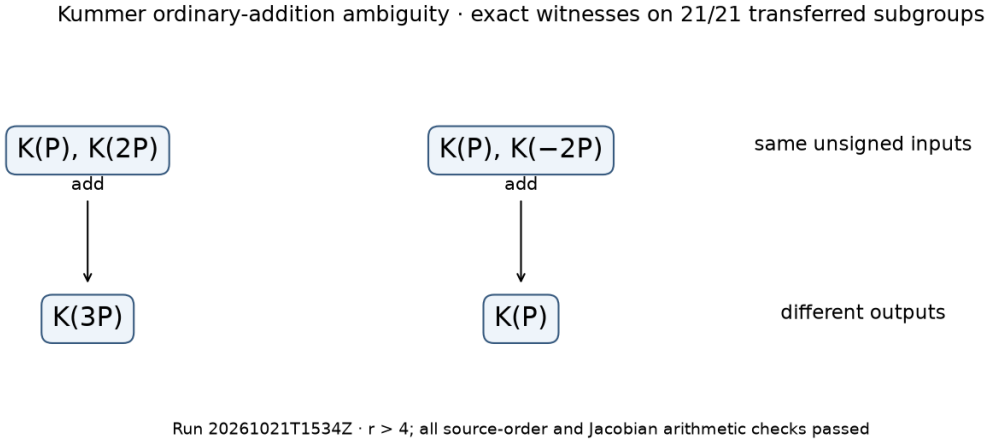


Kummer model and ordinary-addition gates

All 21 transferred quintics have five distinct finite rational Weierstrass roots. Exhaustive evaluation over each base field, derivative checks and reconstruction from the linear factors passed on 21/21 models. The six Weierstrass points including infinity are rational, giving full rational two-torsion on these models. This meets the field-of-definition condition for the fast-Kummer construction in https://eprint.iacr.org/2005/314/. The exact roots are in `results/kummer-model-gate.jsonl`; a fresh rerun matched byte-for-byte.



Set $P = \tau(G)$ and $Q = 2P$. The unsigned quotient gives $K(Q) = K(-Q)$, but $K(P+Q) = K(3P)$ and $K(P-Q) = K(P)$ differ for order $r > 4$. The source-order, Jacobian-arithmetic and distinct-output checks passed on 21/21 exact-order models. This rules out binary ordinary addition from two unsigned Kummer inputs for the current arbitrary-jump walk. A differential walk with a known difference and sign/lift recovery has not been implemented or priced.

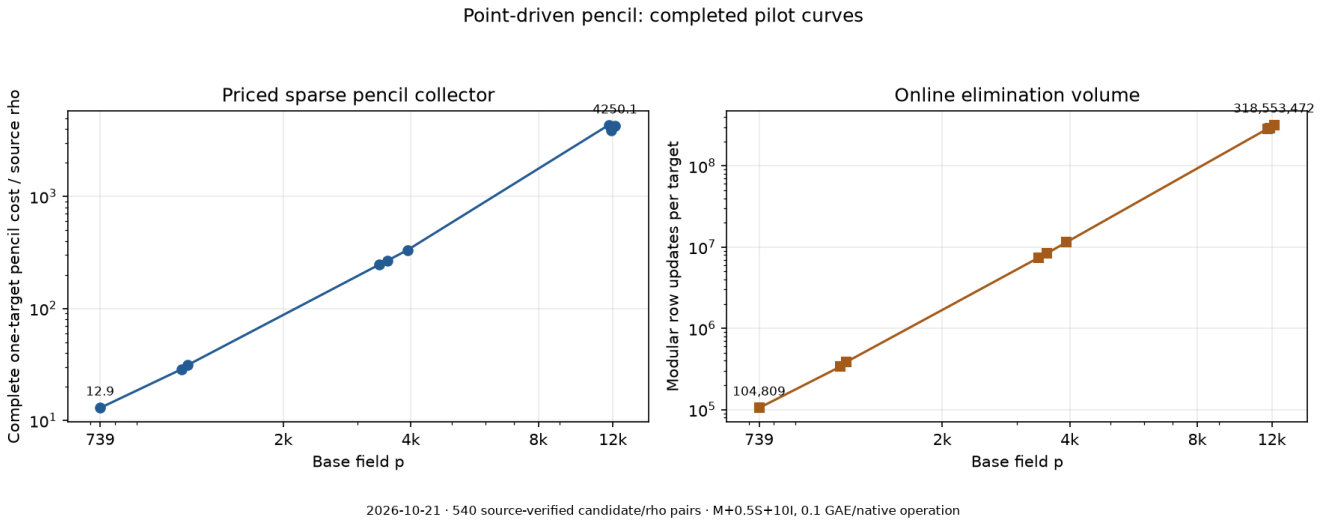


Point-driven pencil pilot

Nine original models, $p = 739, 1151, 1187, 3371, 3527, 3931, 11731, 11887, 12143$, each ran 60 separate targets. All 540 candidate and same-process rho scalars passed source multiplication. The independent validator recomputed all 540 scalar pairs; their replay matched every non-wall field in nine raw files. The direct F739 port also reproduced all 61 prior records after removing wall fields. Full measured setup cost is charged to each cold one-target solve.

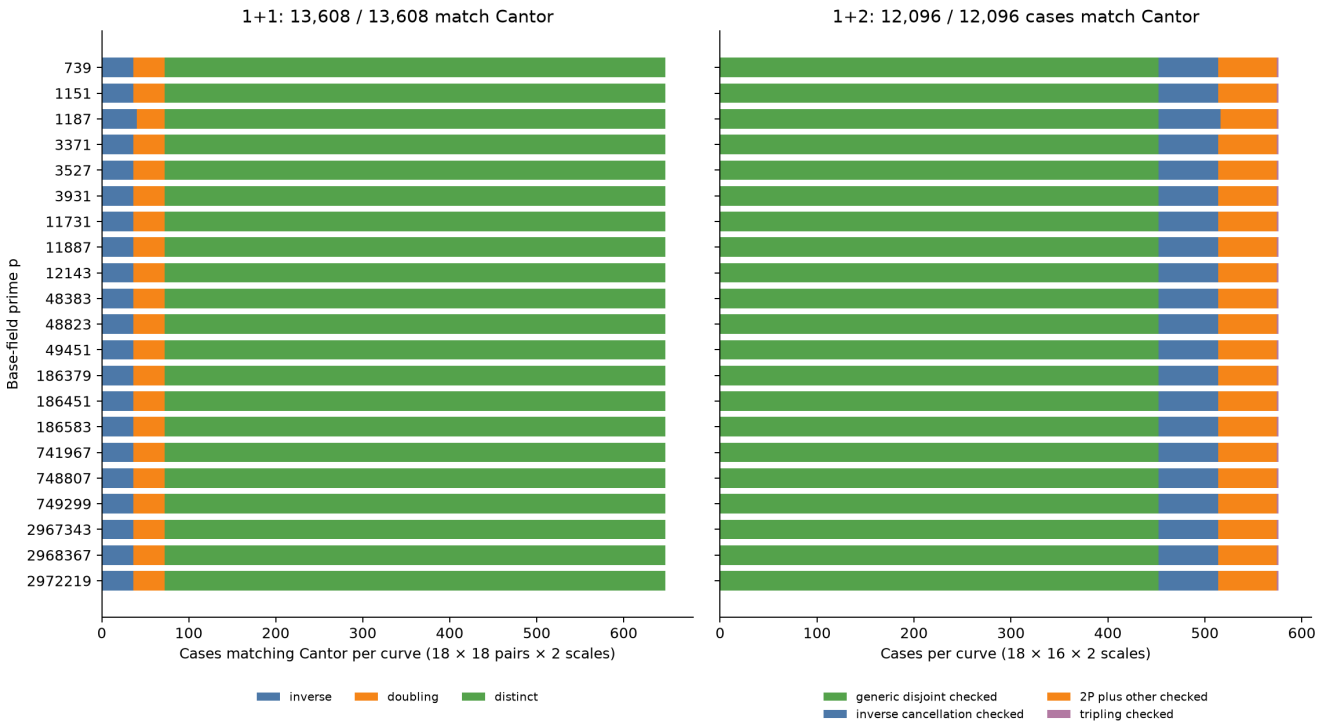
p	Verified	Relations	Modular row updates	Pencil/rho
739	60/60	21,550	6,288,519	12.931178
1,151	60/60	33,703	20,344,734	28.666756
1,187	60/60	34,726	23,112,414	31.080921
3,371	60/60	98,666	442,470,426	246.117110
3,527	60/60	103,256	499,795,794	267.374941
3,931	60/60	114,989	690,243,338	332.187349
11,731	60/60	343,641	17,235,887,587	4,331.361006
11,887	60/60	348,343	17,655,108,982	3,856.238613
12,143	60/60	355,615	19,113,208,311	4,250.094923

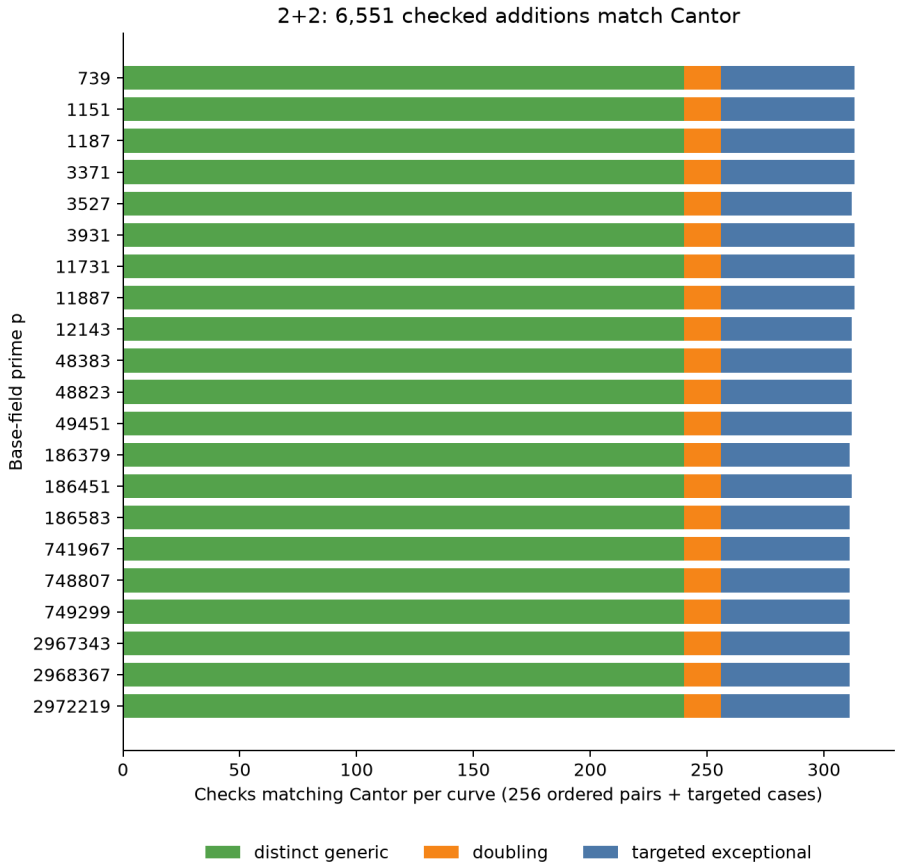
Across the nine models and 36 pricing cells each, 0/324 pencil cells are below matched rho; cell ratios range from 7.870218 to 5,104.745253. The independent validator recomputed all three values in all 324 cells from raw rows. The price is $M + 0.5S + 10I$ plus 0.1 GAE per native logical operation; the independent validator recomputed 29 headline values from all nine models; the analyzer matched fresh rho counters to frozen controls on all nine. The 36-cell price grid and per-target JSONL are in `results/pencil-pilot/`. Upper-size runs are still open. One exploratory 28-bit target at $p = 48{,}383$ also solved with both scalars source-verified. It recorded 20,814,919,424 modular row updates and a 96,957.132823 pencil/rho ratio at the representative price. Its independent replay matched every non-wall field; an independent validator checked both source scalars and all 36 pricing cells. Their ratios range from 26.422722 to 113,636.290374, with 0/36 below rho. See `PENCIL_ONE_TARGET.pdf`; this single target is separate from the seven complete curves.



Complete-law branch gate

Equations (13) through (18) of https://doi.org/10.1007/s13389-024-00351-7 were implemented independently. Across the 21 normalized models and two projective scales, 13,608/13,608 degree-one-pair additions matched Cantor: 760 inverse pairs, 752 doublings and 12,096 distinct-point additions. All 12,096 sampled 1+2 cases matched Cantor: 9,492 generic disjoint, 1,304 inverse-support cancellations, 1,258 repeated-support additions with a distinct remaining point, and 42 triplings. Across 6,551 sampled 2+2 additions, 5,040 were generic, 336 ordinary doublings, 483 degree-drop additions, 336 inverses, 294 shared-support additions, 42 same-polynomial mixed-sign additions, 13 special-point doublings and 7 degree-drop doublings. Every evaluated output matched Cantor. This does not prove exhaustive branch coverage; the full DLP cohort remains unexecuted.


1+1: 13,608 / 13,608 match Cantor
1+2: 12,096 / 12,096 cases match Cantor
Base-field prime p
739
1151
1187
3371
3527
3931
11731
11887
12143
48383
48823
49451
186379
186451
186583
741967
748807
749299
2967343
2968367
2972219
0
100
200
300
400
500
600
Cases matching Cantor per curve (18 × 18 pairs × 2 scales)
Cases per curve (18 × 16 × 2 scales)
inverse
doubling
distinct
generic disjoint checked
inverse cancellation checked
2P plus other checked
tripling checked


2+2: 6,551 checked additions match Cantor
Base-field prime p
739
1151
1187
3371
3527
3931
11731
11887
12143
48383
48823
49451
186379
186451
186583
741967
748807
749299
2967343
2968367
2972219
0
50
100
150
200
250
300
Checks matching Cantor per curve (256 ordered pairs + targeted cases)
distinct generic
doubling
targeted exceptional

Open requirements and evidence

The published complete inversion-free genus-2 formulas are only partly implemented and have not run the 1,260 targets. The point-driven pencil collector has 540 new complete target solves but no full-cohort result here. A differential Kummer collector remains unimplemented. These are open requirements, not negative end-to-end verdicts.

The machine record consists of 35 raw matched JSONL arm files, 6,300 paired candidate/rho target rows, the full 36-cell price grid, summary JSON, 21 Kummer witnesses and 21 model factorizations, nine complete pencil JSONL files with a 36-cell-per-model price grid, and a checksum manifest under `research/four_followups_20261021/`. The native analyzer pairs workload hashes, seeds, caps and source scalars; the full-base arm reproduces the frozen external-law counts and phases. A separate native validator recomputed 15 headline quantities from raw rows; an archive-input rerun was compared byte-for-byte across all 35 files. The plotting script only renders those recorded values. See https://github.com/aburan28/crypto/pull/2304 for the review branch and https://eprint.iacr.org/2024/623.pdf for the separate complete-law path.